Capstone 6 – Quiz: Python Basics (10 Questions)

Choose A, B, C or D. Each question is based on one of the ten programs. The answer key is at the end; the same questions are built into the Colab `grade_quiz()` cell.

1. (P01) What numbers does `range(1, 6)` produce?

- A. 1 to 6
- B. 1 to 5
- C. 0 to 5
- D. 2 to 6

2. (P02) In the countdown program, what is printed right after the number 1?

- A. 0
- B. Liftoff!
- C. -1
- D. Nothing

3. (P03) What does `factorial(0)` return?

- A. 0
- B. 1
- C. An error
- D. None

4. (P04) After `fruits.insert(1, "blueberry")`, where is "blueberry"?

- A. Index 0
- B. Index 1
- C. End of the list
- D. Index 2

5. (P04) What does `fruits[-1]` give?

- A. First item
- B. Last item
- C. An error
- D. Second item

6. (P05) In the comprehension program, what is `squares`?

- A. `[1, 9, 25]`

- B. `[4, 16, 36]`
- C. `[2, 4, 6]`
- D. `[1, 4, 9]`

7. (P06) What does `counts.get(word, 0)` do when `word` is new?

- A. Raises KeyError
- B. Returns 0
- C. Returns None
- D. Adds the word with 1

8. (P07) Why use `with open(...)` for files?

- A. It is faster
- B. It closes the file automatically
- C. It allows Python 2
- D. It encrypts the file

9. (P09) What does the second call `add_item("pear")` print?

- A. `['pear']`
- B. `['apple', 'pear']`
- C. `['apple']`
- D. An error

10. (P10) What does `text[::-1]` do?

- A. Removes the last letter
- B. Reverses the string
- C. Sorts the string
- D. Copies half

Answer key and reasons

Q	Answer	Why
1	B	The end value is excluded
2	B	Loop stops at 0, then Liftoff!
3	B	Empty loop leaves result at 1
4	B	insert places at that index

Q	Answer	Why
5	B	Negative index counts from the end
6	B	Squares of 2, 4, 6
7	B	Default value is 0
8	B	`with` closes the file
9	B	Default list is shared between calls
10	B	Step -1 walks backwards

Scoring: 9–10 strong, 7–8 good, 5–6 revise the mini-guide, below 5 repeat the programs.